

Motion-time graphs

1. A battery operated toy dog starts at a point O and moves in a straight line. Its motion is modelled by the velocity-time graph in Figure 1.

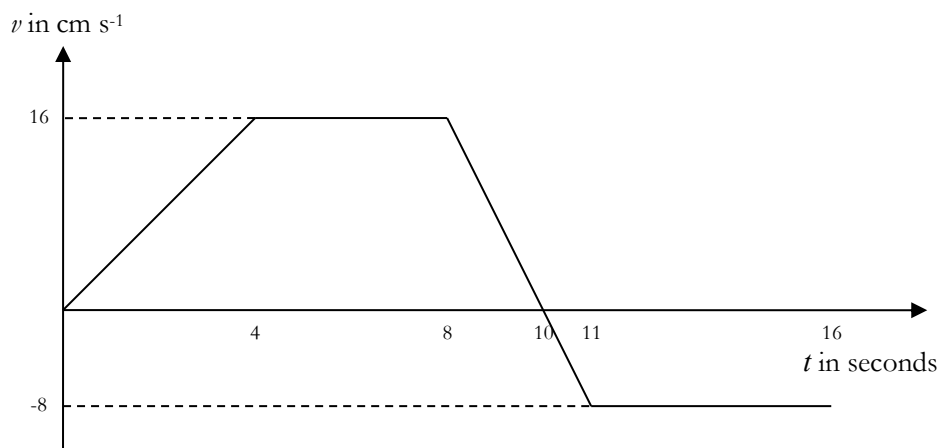


Figure 1

- (a) Calculate the displacement from O of the toy
- after 10 seconds
 - after 16 seconds
- (b) Write down expressions for the velocity of the toy at time t seconds in the intervals $0 \leq t \leq 4$ and $4 \leq t \leq 8$.
- (c) Find the displacement from O of the toy after 1, 2, 3 and 3.5 seconds. In general, find the displacement from O at time t seconds in the interval $0 \leq t \leq 4$.
- (d) Find an expression for the displacement of the toy from O at time t seconds, where $4 \leq t \leq 8$.
2. A bullet is fired from a point O into a sand bank and moves in a straight line. Its motion is modelled by the velocity-time graph in Figure 2.

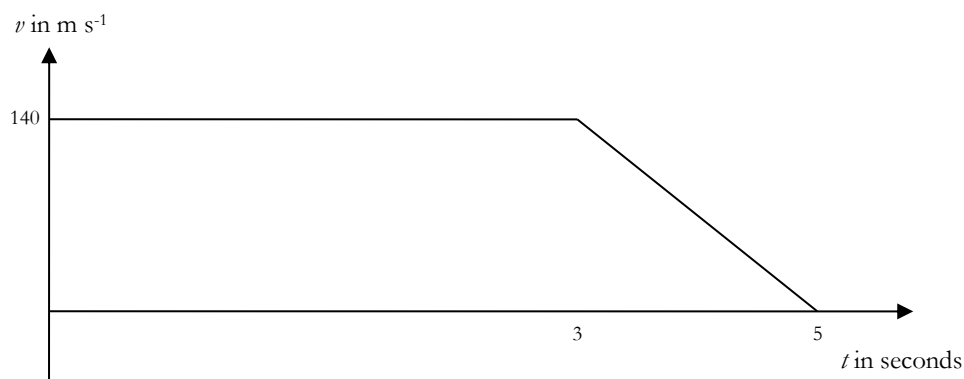


Figure 2

- (a) How far is the sand bank from O?
- (b) How far into the sand bank does the bullet penetrate?
- (c) Write down expressions for the velocity of the bullet at time t seconds in the intervals $0 \leq t \leq 3$ and $3 \leq t \leq 5$.

3. A bus is travelling at a constant speed of 10 m s^{-1} . A car starts moving just as the bus passes it and accelerates to 15 m s^{-1} in 20 seconds. 2 minutes later the car has brake to stop at a set of traffic lights whereas the bus can go straight through in the bus lane. Their motion is modelled by the velocity-time graph in Figure 3.

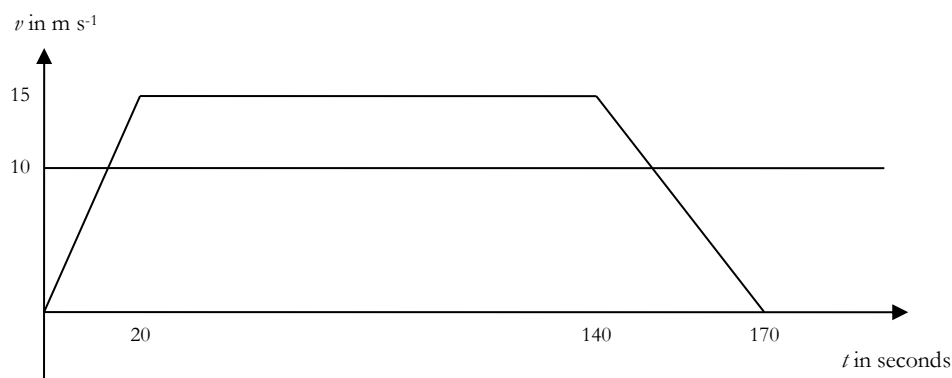
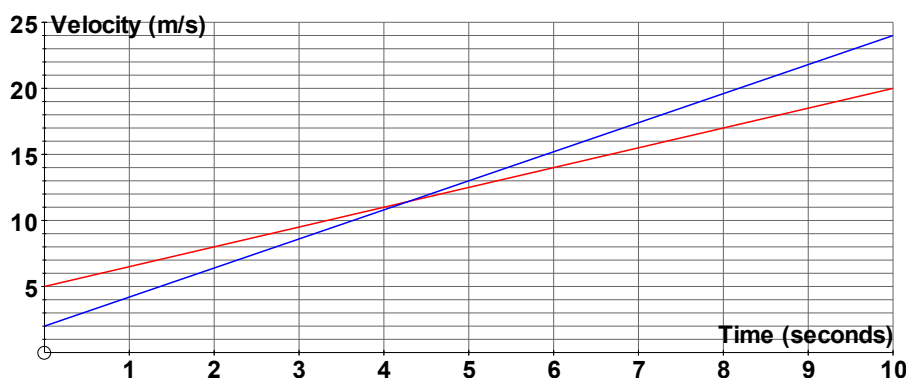


Figure 3

- How far are the traffic lights from the cars starting point?
- How far does the bus travel in the first 170 seconds?
- When does the bus pass the traffic lights?
- How far in front of the bus is the car at 140 s?

4. A car passes a road sign whilst travelling at 2 m s^{-1} . At the same time a second car in a different lane passes the same sign travelling at 5 m s^{-1} in the same direction. The velocity time graphs of the cars are shown below.



- Which car is going faster at
 - $t = 2$?
 - $t = 8$?
- Use the graph to calculate the acceleration of each car.
- Which car initially gets away from the other? Does the other car eventually catch up? If so, at what time does this happen?